🔑 **Answer Key — Teacher Copy · Lesson 6-7**

MSTAR PRACTICE WORKSHEET

Lesson 6-7: Find Factors and Multiples

Grade 6 Mathematics · Standard 6.NOS.4

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

Which list shows all the factors of 12?

- **(A) 1, 2, 3, 4, 6, 12**
- (B) 1, 12
- (C) 2, 3, 4, 6
- (D) 12, 24, 36, 48

A factor divides 12 evenly. Every one of 1, 2, 3, 4, 6, and 12 does, and nothing else does.

Question 2 SELECTED RESPONSE — CORRECT: C

Which list shows the first four multiples of 8?

- (A) 1, 2, 4, 8
- (B) 2, 4, 6, 8
- **(C) 8, 16, 24, 32**
- (D) 8, 9, 10, 11

Multiples come from multiplying by 1, 2, 3, 4: 8, 16, 24, and 32 — the numbers you say skip-counting by 8.

Question 3 SELECTED RESPONSE — CORRECT: C

What is the greatest common factor of 12 and 18?

A 2

B 3

C 6

D 36

> Factors of 12: 1, 2, 3, 4, 6, 12. Factors of 18: 1, 2, 3, 6, 9, 18. They share 1, 2, 3, and 6 — the greatest is 6.

Question 4 SELECTED RESPONSE — CORRECT: C

What is the least common multiple of 4 and 6?

A 2

B 10

C 12

D 24

> Multiples of 4: 4, 8, 12 ...; multiples of 6: 6, 12 .... The smallest they share is 12.

Question 5 SELECTED RESPONSE — CORRECT: C

Using prime factors, $48 = 2 \times 2 \times 2 \times 2 \times 3$ and $36 = 2 \times 2 \times 3 \times 3$. What is their GCF?

A 2, the only prime they share

B 6, from one 2 and one 3

C 12, from the shared $2 \times 2 \times 3$

D 144, from multiplying all the primes

> Take only the primes both numbers have: two 2s and one 3. Multiplying them gives $2 \times 2 \times 3 = 12$, matching the answer from the factor lists.

Question 6 SELECTED RESPONSE — CORRECT: C

Ayasha has painting every 8 weeks and pottery every 6 weeks, both this week. When do they next fall in the same week?

- (A) In 2 weeks
- (B) In 14 weeks
- (C) In 24 weeks
- (D) In 48 weeks

Multiples of 8: 8, 16, 24 ...; multiples of 6: 6, 12, 18, 24 .... The first week they share is week 24.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

What is the greatest common factor of 18 and 24?

- (A) 2
- (B) 3
- (C) 6
- (D) 72

> The factors of 18 are 1, 2, 3, 6, 9, and 18. The factors of 24 are 1, 2, 3, 4, 6, 8, 12, and 24. The shared factors are 1, 2, 3, and 6, and the largest is 6.

- **A:** 2 does divide both, but it is not the greatest one. Keep checking upward and 6 divides both too.
- **B:** 3 does divide both, but it is not the greatest one. 6 divides 18 and 24 as well.
- **D:** That is a common MULTIPLE, and it is larger than both numbers. A factor divides into them.

Part B — correct answer: A

Which reasoning shows why your answer to Part A is correct?

- (A) List the factors of each number, find the ones they share, which are 1, 2, 3, and 6, and take the largest shared factor, 6.
- (B) Stop at the first prime that divides both numbers, which is 3.
- (C) Find the smallest multiple that 18 and 24 share, which is 72, and use it as the greatest common factor.
- (D) Both 18 and 24 are even numbers, so the greatest shared factor is 2.

> The greatest common factor is the largest factor in both factor lists. The number 72 is the least common multiple, and stopping at 3 or 2 skips the larger shared factor 6.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about the numbers 12 and 20.

- **A** 6 is a factor of 12.
- **B** The greatest common factor of 12 and 20 is 4.
- **C** The least common multiple of 12 and 20 is 60.
- **D** The greatest common factor of 12 and 20 is 60.
- **E** 40 is a multiple of 20.
- **F** 5 is a factor of 12.

> The factors of 12 are 1, 2, 3, 4, 6, and 12, so 6 is a factor and 5 is not. The factors of 20 are 1, 2, 4, 5, 10, and 20, so the greatest common factor is 4. Multiples of 12 are 12, 24, 36, 48, 60 and multiples of 20 are 20, 40, 60, so the least common multiple is 60. Since $20 \times 2 = 40$, the number 40 is a multiple of 20.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Reginald is having hot dogs for 24 guests and wants the same number of hot dogs and buns, with none left over. Hot dogs come in packs of 10 and buns in packs of 8. How many packs of each should he buy, and how many will he have? Explain which tool you used and why.

> **Model answer:** This is a least-common-multiple problem: the pack sizes repeat, and he needs the first count both can reach exactly. Multiples of 10: 10, 20, 30, 40 ...; multiples of 8: 8, 16, 24, 32, 40 .... The first number in both lists is 40, so the LCM of 10 and 8 is 40. That means 4 packs of hot dogs ($4 \times 10 = 40$) and 5 packs of buns ($5 \times 8 = 40$), giving 40 of each with nothing left over. It is a multiple problem rather than a factor one because nothing is being split into groups — he is counting packs forward until the two totals match. He will have 40 of each for 24 guests, which is more than enough; buying fewer packs would leave hot dogs or buns unmatched.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.
